

Segmentation of the Left Ventricle Using Active Contour Method with Gradient Vector Flow Forces in Short-Axis MRI

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Abstract. In this paper a left ventricle segmentation approach in short-axis MRI is proposed. It is based on an active contour method and gradient vector flow field forces. Firstly, algorithm delineates endocardium using active contour method approach assisted by gradient vector flow field forces. After that, the epicardium is outlined by proposed divergence rays method and corrected by Fourier descriptors to smoothen an epicardium curve.

An algorithm has been tested on eight healthy patients and compared to a manual delineation of endo- and epicardium boundaries. Validity of an algorithm is checked by linear regression analysis, correlation coefficients, and RSME errors. Sample Pearson product-moment correlation coefficients between automatic and manual delineation are $r_{ENDO} = 0.95$ and $r_{EPI} = 0.86$. The coefficients of determination and RMSEs are $R_{ENDO}^2 = 0.9$, $R_{EPI}^2 = 0.74$ and $RMSE_{ENDO} = 5.303\text{ ml}$, $RMSE_{EPI} = 21.973\text{ ml}$, respectively. These experiments confirm accuracy and robustness of the proposed approach.

Keywords: left ventricle, image segmentation, active contour, gradient vector flow, Fourier descriptors.

1 Introduction

The aim of this article is to present a left ventricle (LV) segmentation method in short-axis MRI images based on an active contour method (ACM) preceded by gradient vector flow (GVF) field forces. A proposed segmentation method extracts a left ventricle muscle at a fixed slice and single cardiac phase. Left ventricle areas are commonly delineated manually by cardiologists, which is highly time consuming task. In most cases, cardiac MRI images contain various inconveniences, a few of which are shown in Fig. 1. The most frequently encountered ones are non-continuous myocardium area, papillary muscles in ventricular cavity, and fuzzy borders between myocardium and ventricular cavity as well as chest cavity.

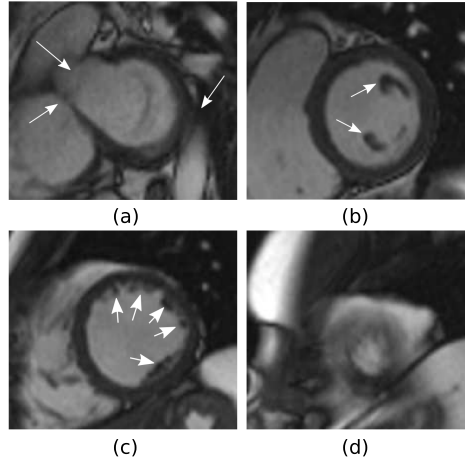


Fig. 1. Demonstration of few complications in a segmentation process. Picture (a) consists of non-continuous endocardium border. Also, there is no clear border of epicardium due to basal cut plane (pointed out by arrows). In picture (b), there are capillary muscles within a ventricular cavity area. Third image (c) shows a cavity, which is disturbed by other muscles connected to a myocardium wall. Last case (d) presents ambiguous area of myocardium due to apex tranverse plane.

Segmentation of the left ventricle is fundamental and inherent process of further cardiac function analysis e.g., strain and strain rate analysis, wall thickness calculation, ejection fraction assessment, peak ejection and computation of filling rate as well as three-dimensional visualisation [2][5].

Until recently, many different methods of a segmentation of cine cardiac MRI have been proposed including level set approach [11], graph cuts [19], Markov random field model [18], iterative thresholding and an active contour model [8], dynamic programming and fuzzy connectedness [7] and as well as knowledge-based approaches such 3-D active shape model [17].

The prior knowledge-based models are characterized by good accuracy in segmentation process but sample shapes of the organs should be prepared before applying the methods. Also, there are few articles, which present fully automatic approach to segmentation of left (right) ventricle. For example, in [2] fully automatic left and right ventricle segmentation is presented, but it is based on modern SSFP scan protocol, which, in fact, indicates improved quality of images. A good review of most recent advances in segmentation algorithms in short-axis cine MRI are found in [12].

This article focuses on application of the parametric active contour approach with gradient vector flow forces and it is organized as follows. In the next section, an active contour method and gradient vector flow is recalled. In Sect. 3, myocardial wall segmentation algorithm is proposed as well as some improvements of an active contour method. In the last section, results are shown and concluding remarks are drawn.

2 An Active Contour Method with Gradient Vector Flow Forces

An active contour method (also called *snakes*) proposed by Kass et. al. [6] has significant influence on the development direction of computer vision algorithms, and on image segmentation in particular. Unfortunately, basic form of active contour approach is characterized by slow (or none) curve convergence to object's boundary and difficulties with fitting curve to narrow concavities [20]. There is other major inconvenience, notably ACM approach is unable to extract multiple objects on the scene. Many improvements have been proposed to accelerate convergence of the curve such as the balloons model [3], gradient vector flow forces [20], and vector field convolution approach [9]. Also, specific initialization methods of the curve have been developed such as quasi-automatic initialization [15] and Poisson inverse gradient [10].

This universal approach with some improvements have been used in many types of medical image segmentations such as lungs [14], liver cancer [16], retinal vessels [1], and naturally a left ventricle of the heart [8]. Below, a short description of active contour approach is recalled.

Let $\Omega \subset \mathbb{R}^2$ be a rectangular bounded set and $\psi: [0, 1] \rightarrow \Omega$ be the parametric curve defined as $\psi(s) = (x(s), y(s))$. Total energy of deformable curve $\psi(s)$ is given by:

$$E_{snake} = \int_0^1 E_{int}(\psi(s)) + E_{ext}(\psi(s)) ds \quad (1)$$

where E_{int} and E_{ext} denotes internal and external energy of the curve, respectively. An internal energy is defined as follows:

$$E_{int} = (E_{int} \circ \psi)(s) = \frac{1}{2} \left(\alpha \left| \frac{\partial \psi}{\partial s} \right|^2 + \beta \left| \frac{\partial^2 \psi}{\partial s^2} \right|^2 \right) \quad (2)$$

where α is tension weighting parameter and β is rigidity parameter. First-order partial derivative treats snake as a membrane, while second-order makes it act like a thin-plate [6]. An external energy is based on the image content and could be delineated in many ways, e.g.,

$$E_{ext} = -|\nabla I(x, y)|^2 \quad (3)$$

where $I: \Omega \rightarrow \mathbb{R}$ is a two-dimensional gray-level image of continuous variables (x, y) . To minimize energy (1), according to calculus of variations, curve $\psi(s)$ must satisfy Euler-Lagrange equation:

$$\frac{\partial}{\partial s} \left(\alpha \frac{\partial \psi}{\partial s} \right) - \frac{\partial^2}{\partial s^2} \left(\beta \frac{\partial^2 \psi}{\partial s^2} \right) - \nabla E_{ext} = 0 \quad (4)$$

$$-\nabla E_{ext} = \begin{bmatrix} -\frac{\partial E_{ext}}{\partial x} \\ -\frac{\partial E_{ext}}{\partial y} \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \end{bmatrix} \quad (5)$$

After discretizing vector equation (4), the method takes an iterative form as follows:

$$\begin{cases} X^\tau = (I_n + \gamma A)^{-1} (X^{\tau-1} + \gamma f_x(X^{\tau-1}, Y^{\tau-1})) \\ Y^\tau = (I_n + \gamma A)^{-1} (Y^{\tau-1} + \gamma f_y(X^{\tau-1}, Y^{\tau-1})) \end{cases} \quad (6)$$

where I_n is identity matrix of $n \times n$ size, A is almost pentadiagonal symmetric matrix of $n \times n$ size, γ is time step, and discrete equivalent of $\psi(s) = (x(s), y(s))$ is $\underline{\psi}(i) = (X_i, Y_i)$. Markers (also called snaxels) in iteration τ are represented by vectors:

$$\begin{cases} X^\tau = [X_0^\tau, \dots, X_{n-1}^\tau]^T \\ Y^\tau = [Y_0^\tau, \dots, Y_{n-1}^\tau]^T \end{cases} \quad (7)$$

In the following article, symbol $\underline{\psi}(i^\tau)$ means curve $\underline{\psi}(i)$ in iteration τ .

2.1 Gradient Vector Flow

As mentioned in the previous paragraph, slow (or none) curve convergence is a drawback of an active contour method, as it can be seen on the smoothed edge image.

To solve this inconvenience, a gradient vector flow forces was proposed [20]. The aim of this approach is to set up a gradient for each image pixel, even if the gradient itself has a small or even zero magnitude. New gradient field is calculated in an iterative way.

The method assumes that initial forces are equal [20]:

$$F_{int} = F_{ext} \quad (8)$$

In a basic active contour method, an equilibrium of forces is reached along with a minimum of total energy (1) [6]. New GVF field $V: \Omega \rightarrow \mathbb{R}^2$ is defined as follows:

$$V(x, y) = [u(x, y), v(x, y)] \quad (9)$$

In this case, following energy functional is being minimized [20]:

$$E_{snake}^{GVF} = \iint_{\Omega} \mu (u_x^2 + u_y^2 + v_x^2 + v_y^2) + (f_x^2 + f_y^2) ((u - f_x)^2 + (v - f_y)^2) dx dy \quad (10)$$

where μ is non-negative parameter controlling field smoothness (the noisier image, the greater value should be) and $f: \Omega \rightarrow \mathbb{R}^2$ is the edge force derived from an image $I(x, y)$.

3 Myocardial Wall Segmentation in Short-Axis

An algorithm for segmentation of left ventricle in short-axis MRI consists of two stages related to each other. Firstly, internal border of myocardium (endocardium) is being delineated, basing on slightly improved active contour method

with gradient vector flow forces. As a result of this stage, a closed curve which defines an area between left ventricle cavity and heart muscle is obtained. Second stage delineates an external border of myocardium (epicardium) and is based on a proposed algorithm called *divergence rays method*, supported by Fourier descriptors.

3.1 The Endocardium Contour Delineation

The endocardium delineation algorithm is organized as follows:

1. Low-pass filtering.
2. Edge detection on a filtered image.
3. Calculation of gradient vector flow forces.
4. Slightly improved active contour procedure.

Each stage is explained in following paragraphs.

Low-pass Filtering

At first, image of left ventricle is filtered by low-pass filter $G_\sigma(x, y)$ (49 coefficients) of Gaussian characteristics with standard deviation $\sigma = 0.5$.

Edge Detection on a Filtered Image

A multiple algorithms, eg., Sobel, Prewitt, Kirsch and Robinson gradient as well as non-linear combining filters, have been tested. The results of most of them are not satisfying from active contour method view-point. All of them were rejected, because they generate non-continuous edges, double edges, and detect isolated points, which do not belong to a real endocardium boundary. The best results were achieved by Canny operator, which was finally used.

This edge image is a basis to define vector field with gradient vector flow method, which is used in following to attract a curve to the boundaries of the tissue.

Calculation of Gradient Vector Flow Forces

The vector field is generated by GVF algorithm in iterative way, basing on an edge image. Parameters of this process are defined as follows: field smoothness $\mu = 0.2$, iteration number – 50.

The GVF field is then transformed by non-linear combination of $u(x, y)$ and $v(x, y)$ vectors to new vector of external forces $F_{ext}^{GVF} = [f_x^{GVF}, f_y^{GVF}]$, according to equations:

$$f_x^{GVF} = \frac{u}{\sqrt{u^2 + v^2} + 10^{-10}} \quad (11)$$

$$f_y^{GVF} = \frac{v}{\sqrt{u^2 + v^2} + 10^{-10}} \quad (12)$$

Slightly Improved Active Contour Procedure

An active contour method is applied to fit the curve to the endocardium boundary. The algorithm, which was mentioned in Sect. 2 was slightly modified. However, earlier an active contour method have to be initialized, for example as

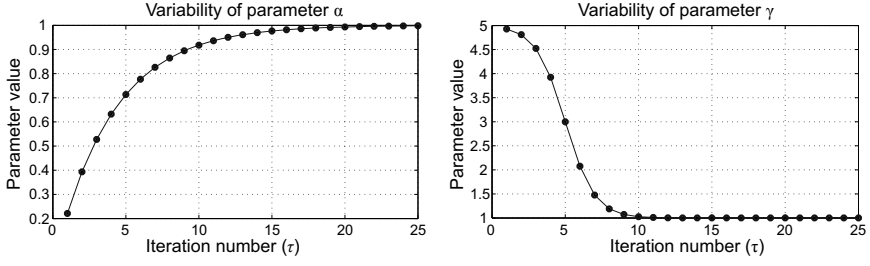


Fig. 2. Values of α and γ parameters as the function of an iteration number

a circle: circle radius – 50, initial points number – 25, points distributed evenly every $\frac{2\pi}{35} \approx 10.3^\circ$.

Parameters α , β and γ are generally constants with values picked accordingly to specific situations. However, parameters can be variable, which allows curve to fit properly to endocardium boundary. Variability of these parameters is proposed as follows (Fig. 2):

$$\alpha^\tau = \frac{1}{10} \left(1 - \exp \left(-\frac{1}{4}\tau \right) \right) \quad (13)$$

$$\gamma^\tau = \frac{5 \exp(-(\tau + 5)) + 1}{\exp(-(\tau + 5)) + 1} \quad (14)$$

Growth of α parameter as a function of a current iteration number τ is associated with achieving a flexibility of the curve, while fitting it to narrower fissures. Wherein α parameter is set too high, a curve might start shrinking or converging to the wrong shape. The purpose of making γ parameter variable is to attract a curve to the boundary as fast as possible. Reduction of γ parameter is associated with preventing a curve oscillations around the real boundary. The β parameter is constant ($\beta = 10$).

In a process of fitting a curve many unexpected situations can occur, especially when the curve $\underline{\psi}(i)$ cannot propagate because of few markers existing in local areas (or too many of these markers). To overcome such problems, the curve is resampled after every iteration, according to the scheme shown in Algorithm 1 scheme. Resampling algorithm assumes that:

$$(X_{-1}^\tau, X_{-1}^\tau) = (X_{n-1}^\tau, X_{n-1}^\tau) \quad (15)$$

The distance between two points is calculated according to ℓ^2 -norm:

$$\|X\|_2 = \sqrt{\sum_{i=0}^n |X_i|^2} \quad (16)$$

Fitting process by an active contour method contains an additional stop condition. After each iteration, an algorithm checks the value of difference $\varepsilon_{\underline{\psi}}^\tau$ between

Algorithm 1. A curve $\underline{\psi}(i^\tau)$ resampling scheme

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1:  $p \leftarrow 0, i \leftarrow 0$ 
2: for all snaxel  $i$  of curve  $\underline{\psi}(i^\tau)$  do
3:   if  $\|X_i^\tau - X_{i-1}^\tau\| + \|Y_i^\tau - Y_{i-1}^\tau\| \geq 8$  then
4:      $X_p'^\tau \leftarrow X_{i-1}^\tau, Y_p'^\tau \leftarrow Y_{i-1}^\tau$ 
5:      $X_{p+1}'^\tau \leftarrow \frac{X_i^\tau + X_{i+1}^\tau}{2}, Y_{p+1}'^\tau \leftarrow \frac{Y_i^\tau + Y_{i+1}^\tau}{2}$ 
6:      $X_{p+2}'^\tau \leftarrow X_i^\tau, Y_{p+2}'^\tau \leftarrow Y_i^\tau$ 
7:      $p \leftarrow p + 3, i \leftarrow i + 1$ 
8:   else if  $\|X_i^\tau - X_{i-1}^\tau\| + \|Y_i^\tau - Y_{i-1}^\tau\| \leq 2$  then
9:      $X_p'^\tau \leftarrow X_i^\tau, Y_p'^\tau \leftarrow Y_i^\tau$ 
10:     $p \leftarrow p + 1, i \leftarrow i + 2$ 
11:   else
12:      $X_p'^\tau \leftarrow X_i^\tau, Y_p'^\tau \leftarrow Y_i^\tau$ 
13:     $p \leftarrow p + 1, i \leftarrow i + 1$ 
14:   end if
15: end for

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a new curve and old one (before resampling), which are respectively represented by (X_i^τ, X_i^τ) and $(X_i^{\tau-1}, X_i^{\tau-1})$. If difference $\varepsilon_{\underline{\psi}}^\tau$ in τ -iteration is less than defined tolerance $\varepsilon_{\underline{\psi}}^{TOL}$, an algorithm stops after performing no less than minimal numbers of iterations $\tau_{\underline{\psi}}^{MIN}$. Otherwise, if the difference is greater than or equal to set tolerance, algorithm does not stop, even if minimal number of iterations has been performed.

The direct reason on such approach is the existence of a possibility of transitional settling of a curve in local minimum, or having a curve that oscillates after reaching the borderline. Selection of these parameters has been done in an empirical way as follows: $\tau_{\underline{\psi}}^{MIN} = 30$ and $\varepsilon_{\underline{\psi}}^\tau = 5$.

Difference $\varepsilon_{\underline{\psi}}^\tau$ is calculated according to the formula:

$$\varepsilon_{\underline{\psi}}^\tau = \max_i (\|X_i^\tau - X_i^{\tau-1}\|_2 + \|Y_i^\tau - Y_i^{\tau-1}\|_2) \quad (17)$$

3.2 An Epicardium Contour Delineation

The outline of an endocardium, which was obtained in the previous step, is now basis for an algorithm for delineation of a contour of epicardium. Proposed approach composes of two steps:

1. A *divergence rays* algorithm is applied in order to find preliminary epicardium markers [13].
2. Fourier descriptors are used to smoothen epicardium curve (represented by markers) [4].

Divergence Rays Algorithm

The purpose of *divergence rays* algorithm is to make a preliminary estimation of the outline of an epicardium.

In first stage of the algorithm, pencil of rays is calculated, basing on a middle point (C) of endocardium markers (P_i) up to S_i . In this case, length of each ray ($n = 0, 1, \dots, n-1$) is limited to:

$$|CS_i| = 2|CP_i| \quad (18)$$

This assumption has been made due to anatomical structure of the left ventricle and the chest cavity.

Then, variability of the luminance along each ray (excluding points between C and P_i) is extracted from the edge image, which was created in endocardium contour delineation step. Finally, algorithm is looking for an intersection between each ray and the edge of the epicardium based on the approach proposed in [13].

Fourier Descriptors

Second step of the epicardium contour delineation algorithm is an application of Fourier descriptors. The purpose of this step is to smoothen rough curve, which is a result of the previous step. Each point of the curve $R'_i = (rx'_i, ry'_i)$ is being transformed into a complex number as follows [4]:

$$\mathbf{R}' = [rx'_0 + jry'_0, \dots, rx'_{n-1} + jry'_{n-1}]^T \quad (19)$$

where $j^2 = -1$ is an imaginary unit.

Then, discrete Fourier transform is applied to the vector $\mathbf{R}'$:

$$\hat{\mathbf{R}}'_k = \sum_{i=0}^{n-1} \mathbf{R}'_i \exp\left(\frac{-j2\pi ik}{n}\right), \quad k = 0, 1, \dots, n-1 \quad (20)$$

High-frequency coefficients which correspond to unexpected variability of the curve, are then set to zero (in other words, low-pass filter is applied) as follows:

$$\hat{\mathbf{R}}'_k = \begin{cases} \hat{\mathbf{R}}'_k & \text{if } k = 0, \dots, z-1 \\ 0 & \text{if } k = z, \dots, n-1 \end{cases} \quad (21)$$

In the end, inverse Fourier transform is applied to partially zeroed frequency coefficients:

$$\tilde{\mathbf{R}}_i = \frac{1}{n} \sum_{k=0}^{n-1} \hat{\mathbf{R}}'_k \exp\left(\frac{j2\pi ki}{n}\right), \quad i = 0, 1, \dots, n-1 \quad (22)$$

Application of Fourier descriptors results in a smoothening of the curve (Fig. 3). Moreover, points, which look beyond epicardium morphology are corrected. An important fact is that presented approach is invariant with respect to linear transformations (eg., rotation or scaling).

Another research topic is the number of zeroed coefficients z . The most commonly used number of zeroed coefficients is $z = 10$, and such value is applied to the whole dataset.

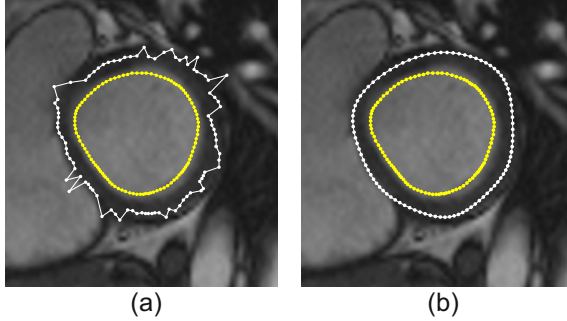


Fig. 3. Results of epicardium contour delineation after corrections in the previous step of the algorithm (a). Visible smoothing of an epicardium curve after Fourier descriptors have been applied ($z = 10$ coefficients set to zero) (b).

4 Results and Concluding Remarks

4.1 Dataset

The data from eight healthy patients (males, aged: 41-61, ± 9.67 standard deviation) has been acquired by Siemens SonataVision 1.5 T scanner (The John Paul II Hospital in Cracow, Poland). Imaging parameters: slice thickness: 6 mm, TE 1.19 – 1.25 ms, TR 39.96 – 41.58 ms, FA 55 – 60, FOV (Field of View) 276 – 340 $\times$ 156 – 192. Each examination has returned 10 slices and 25 imaging frames (cardiac phases).

4.2 Results

At first, because of unclear LV borderlines, basal imaging plane was removed as well as an apex plane. Images have been delineated manually by a radiologist with a specialization in cardiac MRI. Exemplary endocardial and epicardial LV volumes were generated, basing on automatic tracing, are shown in Fig. 4.

Linear regression analysis was performed in order to assess a correlation between automatic and the manual volume of endo- and epicardium. Sample results are shown in Fig. 5. In addition, Pearson product-moment correlation coefficient r and coefficient of determination R^2 have been calculated as well. The coefficient of determination R^2 gives the information how much of a fluctuation of the variable is predictable from another one.

For endocardium and epicardium volumes, Pearson product-moment correlation coefficient and coefficient of determination are respectively $r_{ENDO} = 0.95$, $R^2_{ENDO} = 0.9$ and $r_{EPI} = 0.86$, $R^2_{EPI} = 0.74$. In the first case, this score means there is highly strong correlation between manual and automatic contour delineation of LV endocardium.

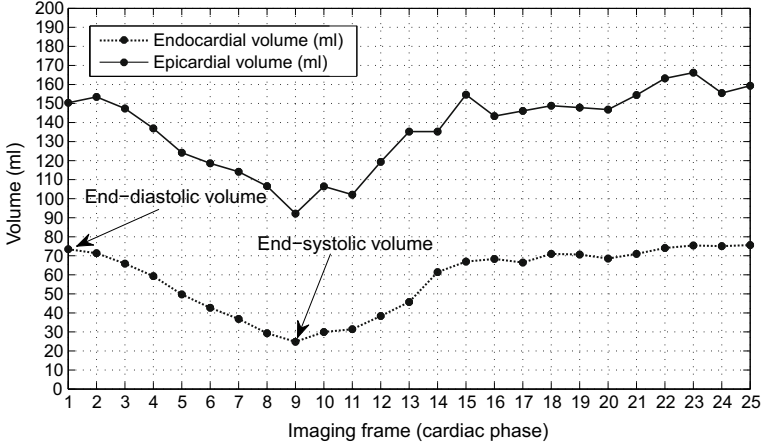


Fig. 4. Endo- and epicardial volume calculations based on proposed algorithm

For each dataset, root mean square error (RMSE) between volumes in all phases has been calculated as well, according to the formula:

$$RMSE = \sqrt{\frac{1}{25} \sum_{m=1}^{25} (VOL_m^{automatic} - VOL_m^{manual})^2} \quad (23)$$

where VOL_m is either endo- or epicardium volume in m -th phase. The endo- and epicardium volumes at each phase are obtained by inside points summation of endo- and epicardium curves, respectively.

Summary of results for sample dataset is included in Table 1. Also, medical parameters as end-diastolic volume (EDV), end-systolic volume (ESV), stroke volume (SV), and ejection fraction (EF) are obtained (Table 2).

Table 1. Statistical parameters comparison for endo- and epicardium volume

statistical parameter	endocardium volume	epicardium volume
r	0.95	0.86
R^2	0.9	0.74
$RMSE$	5.303 ml	21.973 ml

Table 2. Diagnostics parameters obtained from automatic delineation of LV volumes

parameter	parameter calculation	manual	automatic
end-diastolic volume (EDV)	–	84.1 ml	73.46 ml
end-systolic volume (ESV)	–	30.64 ml	24.85 ml
stroke volume (SV)	$SV = EDV - ESV$	53.46 ml	48.61 ml
ejection fraction (EF)	$EF = (SV/EDV) \cdot 100\%$	63.57%	61.17%

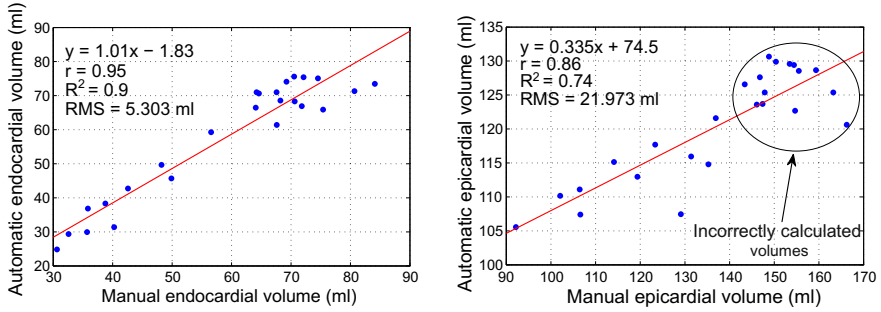


Fig. 5. Comparison of automatic and manual delineation of endo- and epicardium volume by linear regression analysis, Pearson product-moment correlation coefficient r , coefficient of determination R^2 , and root mean square error ($RMSE$)

As it is shown, these analyses confirm accuracy and robustness of automatic endocardium delineation. Statistical analyses have proved that proposed segmentation algorithm works incorrectly for cardiac phases at the end of LV contractions (Fig. 5b, data pointed out by arrow). Segmentation at the beginning of systole and at the end of diastole phase works properly (Fig. 5).

4.3 Conclusion and Remarks

In this article, a segmentation algorithm for endo- end epicardium in short-axis cardiac MRI has been proposed. Eight healthy patients were examined, whose myocardiums have been outlined by a radiologist prior to the evaluation. According to the results presented in a previous section, this algorithm can be successfully used in the endocardium volume segmentation process. Note, however, that some cardiac phases of epicardium LV, which correspond to the end of ejection phase and the beginning of the isovolumic relaxation phase, should still be corrected by a professional. Future work will consider what improvements can be done to an algorithm, when it comes to presented inconveniences with a delineation of an epicardium of the left ventricle.

Acknowledgement. This work is founded by the AGH University of Science and Technology as a research project No. 11.11.120.612.

The author would like to thank R. Banys and T. Miszański-Jamka, from The John Paul II Hospital in Cracow, Poland, for supplying the image data, informative discussions, and commitment to this project.

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